STEP 4 — Poora Pipeline Chalana (Step-by-Step)

Time: ~17 seconds runtime · Is step ka output: 14 charts + saare results files

> Pipeline ke **6 stages** hain. Aap chahein to ek command me sab chalayein (`python run_all.py`) ya ek-ek karke (recommended — report me likhne ke liye har stage ka output samajhna zaroori hai).

4.1 Sabse pehle: venv activate kijiye

```
# Windows PowerShell
.\.venv\Scripts\Activate.ps1

# Mac / Linux
source .venv/bin/activate
```

Prompt me `(.venv)` dikhna chahiye. Yeh **har naye terminal** me karna padta hai.

4.2 Ek-command wala raasta (fast)

```
python run_all.py
```

Poora pipeline chal jayega. Expected output:

```
======================================================================
 CareFlow - Clinical Pathway Process Mining  |  full pipeline
======================================================================

>>> Step 1 - Generate clinical event log
13:11:33 | INFO | careflow.generate_data | events=13664  cases=1500  activities=15

>>> Step 2 - Clean & engineer features
13:11:35 | INFO | careflow.preprocess | raw events=13664 raw cases=1500
13:11:36 | INFO | careflow.preprocess | clean events=13583 cases=1500 variants=14
13:11:36 | INFO | careflow.preprocess | saved -> data/processed/event_log_clean.csv

>>> Step 3 - Process discovery
13:11:40 | INFO | careflow.discover | events=13583 cases=1500
13:11:40 | INFO | careflow.discover | variants=14 top_variant_share=29.9% dfg_edges=28

>>> Step 4 - Insights & visualisation
13:11:44 | INFO | careflow.insights | total cases=1500 cost=INR 25886761
bottleneck=Observation

>>> Step 5 - Conformance checking
13:11:46 | INFO | careflow.conformance | cases=1500 fitness=1.0000 conforming=100.0%

>>> Step 6 - Machine-learning model
```

```
13:12:46 | INFO | careflow.ml | design matrix: 1500 cases | full=19 features | leakage-
free=14 features
13:12:51 | INFO | careflow.ml | [full         ] random_forest   acc=1.000 f1=1.000
auc=1.000
13:12:55 | INFO | careflow.ml | [leakage_free ] gradient_boosting acc=0.997 f1=0.800
auc=0.998
13:13:03 | INFO | careflow.ml | saved production model -> outputs/models/best_model.joblib

============================================================================
 Pipeline finished in 16.5 s
============================================================================
```

4.3 Stage-by-stage wala raasta (recommended — report ke liye)

STAGE 1 — Event log banana / import karna

```
python src/generate_data.py
```

Kya hota hai: 1,500 patients ka 6 mahine ka hospital event log banta hai — 4 clinical pathways, 15 activities, 28 resources, aur jaan-boojh kar daali gayi data-quality galtiyan.

Output files: `data/raw/event_log.csv` (13,664 rows), `data/raw/case_attributes.csv` (1,500 rows)

Verification:

```
python -c "import pandas as pd; d=pd.read_csv('data/raw/event_log.csv'); print(len(d),
d.case_id.nunique(), d.activity.nunique())"
13664 1500 15
```

STAGE 2 — Data cleaning aur feature engineering

```
python src/preprocess.py
```

Kya hota hai (yeh 7 kaam karta hai):

1. Timestamp ko proper datetime me convert karta hai
2. Duplicate events hatata hai (same case + activity + time)
3. Blank `resource` ko `UNASSIGNED` se bharta hai
4. Negative / ghalat numeric values ko 0 kar deta hai
5. Bahut chhote (<3 events) ya bahut bade (>200 events) cases hata deta hai
6. Features banata hai: waiting time, elapsed time, rework flag, shift, weekday, month
7. Har case ka **variant** (activity sequence ki string) likh deta hai

Output: `data/processed/event_log_clean.csv` + `outputs/reports/preprocessing_report.json`

```
{
  "rows_in": 13664,        "cases_in": 1500,
  "duplicates_removed": 81, "missing_resource_filled": 203,
  "rows_out": 13583,       "cases_out": 1500,
  "activities_out": 15,    "resources_out": 28,   "variants_out": 14,
```

```
  "date_min": "2026-03-01 07:26:00", "date_max": "2026-08-31 05:46:50"
}
```

Yeh table report me zaroor daaliye — “data cleaning me kya-kya kiya” wala section issee banta hai.

STAGE 3 — Process discovery (asli pathway dhoondhna)

```
python src/discover_models.py --miner inductive
```

Kya hota hai: Directly-Follows Graph (DFG) banta hai jo batata hai kaunsa step kaunse step ke baad aata hai aur kitni baar. Saath me top-10 variants aur activity statistics.

Output:

File	Kya hai
`outputs/figures/dfg_careflow.png`	Process model ka diagram
`outputs/figures/variants_top10.png`	Top 10 pathway variants ka chart
`outputs/reports/dfg_edges.csv`	28 edges with counts
`outputs/reports/variants.csv`	Variant + case count + %
`outputs/reports/activity_stats.csv`	Per-activity events, wait, cost
`outputs/reports/discovery_summary.json`	Summary metrics
`outputs/models/petri_net_inductive.png`	Petri net (sirf pm4py install ho to)

```
{
  "cases": 1500, "events": 13583, "activities": 15, "dfg_edges": 28,
  "unique_variants": 14, "top_variant_share_pct": 29.87,
  "top_5_variant_share_pct": 80.94,
  "start_activities": [["Registration", 1500]],
  "end_activities":   [["Discharge", 1500]],
  "pm4py_available": false, "miner": "inductive"
}
```

Interpretation (viva me bolein): 14 variants matlab patients 14 alag-alag raste le rahe hain. Top 1 variant 29.87 % cases cover karta hai aur top 5 milakar 80.94 % — yaani zyada tar patients ek “standard” pathway follow karte hain, baaki exceptional cases hain.

Alag miner try karna (agar pm4py install hai):

```
python src/discover_models.py --miner heuristics
python src/discover_models.py --miner alpha
```

STAGE 4 — Analytics aur business insights

```
python src/analyze_insights.py
```

Kya hota hai: 7 tarah ke analysis + 7 charts:

Chart	Kya dikhata hai
`kpi_los_distribution.png`	Patients kitni der hospital me rehte hain (LOS)
`case_volume_trend.png`	Har mahine kitne patients aaye
`bottleneck_waiting.png`	Kis step par sabse zyada wait hota hai
`cost_by_department.png`	Kis department ka kharcha sabse zyada
`resource_workload.png`	Kaunsa staff member sabse busy hai
`hourly_heatmap.png`	Hafte ke kaunse din, kis ghante load zyada
`pathway_comparison.png`	Har pathway ka LOS vs cost

Real results (reference run):

Average length of stay	9.65 hours
Median LOS	6.14 hours
P90 LOS	18.69 hours
Long-stay cases (>24 h)	17 cases (1.13 %)
Biggest bottleneck	**Observation** — 226.61 min average wait
Total waiting time (all cases)	868,869.6 minutes
Total treatment cost	INR 25,886,760.54
Costliest department	Operation Theatre
Average cost per case	INR 17,257.84
Busiest resource	`REC-01` — 1,038 events
Peak load	Friday, 14:00

Pathway-wise:

Pathway	Cases	Avg LOS (h)	Avg cost (INR)
P1 Outpatient Standard	521	4.52	5,150.28
P3 Chronic Followup	400	5.77	7,169.72
P4 ICU Escalation	204	15.15	22,281.61
P2 Emergency Surgical	375	17.94	42,107.01

Output: 7 PNG charts + `outputs/reports/insights.json` + `outputs/reports/insights_summary.md`

STAGE 5 — Conformance checking (kitna deviate ho rahe hain)

```
python src/conformance.py
```

Kya hota hai: har patient ke journey ko discovered model se compare karta hai **token-based replay** se. Jo cheez model me allow nahi hai woh “deviation” ban jaati hai.

Fitness formula (report me likhiye):

```
fitness = 0.5 x (1 - missing / produced) + 0.5 x (1 - remaining / produced)
    produced  = actual me chalne wale steps
    consumed  = model ke hisaab se sahi steps
    missing   = model ke allowed transitions jo trace me nahi mile
    remaining = trace ke end me bache hue model tokens
```


```
{
  "model": "DFG (directly-follows graph)", "min_edge_count": 3,
  "cases_checked": 1500, "model_edges": 28,
  "average_fitness": 1.0,
  "perfectly_fitting_cases_pct": 100.0,
  "fully_conforming_cases_pct": 100.0,
  "cases_with_deviations": 0,
  "top_deviations": []
}
```


Honest interpretation (yeh viva me zaroor boliye): fitness 1.0 aaya kyunki yeh **synthetic data** hai jo fixed pathway templates se bana hai — jitni bhi rework loops hain, woh sab generator ke andar allowed hain, isliye model khud ke data ko 100 % explain kar leta hai. **Asli hospital log** par fitness generally **0.70 – 0.95** aata hai, aur `top_deviations` me “Lab Test → Discharge” jaisi skip-hop patterns dikhte hain. Yeh project ko kamzor nahi banata — methodology wahi hai, sirf data synthetic hai.

Output: `conformance_report.json`, `conformance_by_variant.csv`, `conformance_fitness.png`, `deviation_types.png`

STAGE 6 — Long-stay prediction model

```
python src/ml_model.py
```

Kya hota hai: 3 algorithms (Logistic Regression, Random Forest, Gradient Boosting) train hote hain, 2 feature sets par:

- **full** — 19 features (sab kuch, including duration aggregates)
- **leakage_free** — 14 features (duration-derived columns hata di, kyunki asli hospital me admission ke waqt waiting time pata nahi hota — isse “data leakage” kehte hain)

Data leakage kya hai? Agar aap model ko `total_wait` jaise columns doge jo outcome ke baad bante hain, to model 100 % accuracy dikha dega — lekin asli hospital me aise model ka koi fayda nahi. Isliye **production model leakage-free features par banaya gaya**. Yeh baat report aur presentation me likhne se aapka kaam **professional** lagta hai.

Feature set	Model	Accuracy	F1	ROC-AUC	CV F1
Full (19 feat.)	Logistic Regression	1.000	1.000	1.000	0.727 ± 0.167
	Random Forest	1.000	1.000	1.000	1.000 ± 0.000
	Gradient Boosting	1.000	1.000	1.000	1.000 ± 0.000
Leakage-free (14 feat.)	Logistic Regression	0.993	0.500	0.987	0.080 ± 0.160
	Random Forest	0.990	0.000	0.996	0.000 ± 0.000
	Gradient Boosting ★	**0.997**	**0.800**	**0.998**	0.180 ± 0.223

★ = production model (`outputs/models/best_model.joblib`)

Output: `best_model.joblib`, `model_metadata.json`, `model_metrics.json`, `feature_importance.csv`, `model_roc.png`, `model_confusion_matrix.png`, `feature_importance.png`. Poori detail: `CareFlow_Model_Evaluation_Results.pdf`

4.4 Selective run (kuch stages skip karna)

```
python run_all.py --skip-ml      # ML stage ke bina (fast demo ke liye)
python run_all.py --skip-data    # apna data use karna (naya generate nahi)
```

4.5 Settings badalna — config/config.yaml

Setting	Default	Kya karta hai
`n_cases`	1500	Kitne patients ka data
`los_threshold_hours`	24	“Long stay” ki definition — 48 karke dekhiye
`noise_threshold`	0.20	Inductive Miner me kitni noise ignore karni hai
`min_edge_count`	3	DFG me kam se kam kitni baar edge aa raha ho
`top_variants`	10	Chart me kitne variants dikhane hain
`test_size`	0.20	ML me test data ka hissa (80/20 split)
`models`	3 models	Kaunse algorithms train karne hain

Setting badalne ke baad pipeline dobara chalaiye — results update ho jayenge.

4.6 Poora output ka structure (milne ke baad check kijiye)

```
outputs/
|-- figures/    (14 PNG files)
|   |-- dfg_careflow.png            kpi_los_distribution.png
|   |-- variants_top10.png          case_volume_trend.png
|   |-- bottleneck_waiting.png      cost_by_department.png
|   |-- resource_workload.png       hourly_heatmap.png
```

```
|   |-- pathway_comparison.png      conformance_fitness.png
|   |-- deviation_types.png         model_roc.png
|   |-- model_confusion_matrix.png  feature_importance.png
|-- reports/    (JSON + CSV + MD)
|   |-- preprocessing_report.json   discovery_summary.json
|   |-- insights.json               insights_summary.md
|   |-- conformance_report.json     conformance_by_case.csv
|   |-- conformance_by_variant.csv  model_metrics.json
|   |-- dfg_edges.csv               variants.csv
|   |-- activity_stats.csv          feature_importance.csv
`-- models/
    |-- best_model.joblib
    `-- model_metadata.json
```

```
dir outputs\figures        # Windows
ls -lh outputs/figures     # Mac / Linux
```

4.7 Troubleshooting

Error / problem	Solution
`FileNotFoundError: config/config.yaml`	Aap galat folder me ho. `cd` karke `CareFlow` folder me jaaiye
`ModuleNotFoundError`	venv activate nahi hai → Step 4.1
`KeyError` kisi column ka	Aapke CSV ke column names spec se match nahi kar rahe → Step 3.2
Charts empty aa rahi hain	`python run_all.py` dobara chalaiye, aur `data/processed/` folder check kijiye
ML stage bahut slow	`python run_all.py --skip-ml`
pm4py graphviz error	Ignore kijiye — DFG model ban chuka hai. Ya `pip uninstall pm4py`
Kuch bhi pura nahi chal raha	Outputs folder delete kijiye aur `python run_all.py` dobara

14 charts + 12 result files ban gaye! Ab Step 5 — dashboard aur demonstration.